Stability of a Microscopic Black Hole Reactor

Hawking evaporation and a fixed-medium Bondi model

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Original manuscript: 10 May 2026 | Revised: 22 October 2026

Abstract

Can a fixed surrounding medium balance the mass loss of a microscopic black hole? We combine idealized Bondi accretion with a blackbody approximation to Hawking emission, then analyze the resulting mass equation. We find a single positive balance, but a small change in mass drives the system away from it. An exact implicit solution and numerical trajectories show how the evolution differs near and far from the balance. For an illustrative water-like choice of parameters, the reference black hole evaporates rapidly in the blackbody model, while departures from a nearly balanced state can be slow. The instability is an exact property of the fixed-coefficient equation. Continuum accretion, realistic emission spectra and the response of the medium to the hole's radiation lie outside the model.

1 Introduction and scope

Hawking radiation suggests a theoretical energy source whose luminosity increases as a black hole loses mass [1]. If surrounding matter replenishes that mass, equality between gain and loss might appear to offer a stationary state. The relevant question is whether a small change in mass returns the system to that state.

This study grew from an olympiad problem written for the Kyrgyz Republic national physics competition and used in IPhO-2026 team selection (Day 2, Problem 3, April 2026). Hawking emission and Bondi accretion are established starting points; here we analyze the stability of their idealized mass balance, solve the mass evolution exactly and compute the resulting timescales.

First, we derive the coefficients from standard black hole thermodynamics and an idealized accretion law. We then reduce the mass dynamics to a one-variable, dimensionless equation, solve it exactly and analyze departures from equilibrium. Finally, we restore dimensions and use one numerical example to illustrate the timescales.

We consider an uncharged, nonrotating black hole, a prescribed medium with fixed density and sound speed, and no active control. The mass M and time t use the asymptotic Schwarzschild description. The word "reactor" names the motivating question; construction, confinement, energy collection and net useful power are outside the scope of this paper.

2 Preliminaries

2.1 Hawking temperature and an emission approximation

For a Schwarzschild black hole, the horizon radius r_s and Hawking temperature T_H are

$$r_s = \frac{2GM}{c^2}, \qquad T_H = \frac{\hbar c^3}{8\pi G M k_B}. \tag{1}$$

Simple estimates (a Newtonian escape speed for r_s, the uncertainty principle for T_H) reproduce both scalings but not the coefficient of T_H; Appendix A derives it from the regularity of the Euclidean geometry [2, 3].

To obtain a simple baseline, we approximate the hole as a photon blackbody with horizon area $\mathcal{A}_H = 4\pi r_s^2$. The Stefan–Boltzmann law, with $\sigma = \pi^2 k_B^4/(60\hbar^3 c^2)$, gives

$$P_{\rm bb} = \sigma \mathcal{A}_H T_H^4 = \frac{K}{M^2}, \qquad K = \frac{\hbar c^6}{15360\pi G^2}. \tag{2}$$

Mass-energy conservation defines the evaporation coefficient B:

$$\dot{M}_{\rm evap} = -\frac{B}{M^2}, \qquad B = \frac{K}{c^2} = \frac{\hbar c^4}{15360\pi G^2}.$$

These expressions give $K = 3.56 \times 10^{32}\,\mathrm{W\,kg^2}$ and $B = 3.96 \times 10^{15}\,\mathrm{kg^3\,s^{-1}}$. At the illustrative mass $M_0 = 10^7\,\mathrm{kg}$, $T_H = 1.23 \times 10^{16}\,\mathrm{K}$ and $P_{\rm bb} = 3.56 \times 10^{18}\,\mathrm{W}$.

Equation (1) gives the semiclassical temperature, whereas Eq. (2) gives a simplified luminosity. Actual spectra depend on frequency- and spin-dependent transmission probabilities (greybody factors) and on the particle species accessible at the relevant temperature [4, 5]. Here $k_B T_H \simeq 1.06\,\mathrm{TeV}$, where many particle species are emitted, so the photon blackbody underestimates the true luminosity. All luminosities and lifetimes quoted below refer to this baseline.

2.2 Bondi scaling and its coefficient

For steady spherical accretion of a polytropic gas at rest far from the object, the Bondi formula is [6]

$$\dot{M}_{\rm acc} = 4\pi\lambda\rho_\infty \frac{G^2M^2}{c_\infty^3} = AM^2, \qquad A = \frac{4\pi\lambda\rho_\infty G^2}{c_\infty^3}.$$

Here ρ_∞ and c_∞ are the distant density and sound speed, and λ depends on the flow and equation of state. For the limiting polytropic solution with adiabatic index $\gamma = 5/3$, $\lambda = 1/4$.

We use the original water-like reference values $\rho_\infty = 10^3\,\mathrm{kg\,m^{-3}}$, $c_\infty = 1500\,\mathrm{m\,s^{-1}}$ and $\lambda = 1/4$. They give $A = 4.15 \times 10^{-27}\,\mathrm{kg^{-1}\,s^{-1}}$. A liquid is not the polytropic gas assumed in the Bondi derivation, so these values define a formal example rather than a validated accretion model for water. The characteristic Bondi scale, in our convention, is

$$r_B = \frac{GM}{c_\infty^2}, \qquad r_B(M_0) = 2.97 \times 10^{-10}\,\mathrm{m}.$$

At M_0 this is comparable to molecular scales. The example therefore lacks the separation of scales required for continuum flow, and the hole's radiation would also heat the medium, which the model holds fixed.

2.3 Combined mass dynamics

With A and B held positive and constant, the model becomes

$$\dot{M} = f(M) = AM^2 - \frac{B}{M^2}, \qquad M > 0. \tag{3}$$

At M_0,

$$AM_0^2 = 4.15 \times 10^{-13}\,\mathrm{kg\,s^{-1}}, \qquad \frac{B}{M_0^2} = 39.6\,\mathrm{kg\,s^{-1}}.$$

The accretion-to-evaporation ratio is 1.05×10^{-14}. Accretion is negligible at M_0; the two terms balance only at a much larger mass (Sec. 3.1).

3 Stability and an exact solution

3.1 A unique, unstable equilibrium

Setting $f(M) = 0$ in Eq. (3) gives one positive balance mass,

$$M_{\rm crit} = \left(\frac{B}{A}\right)^{1/4}. \tag{4}$$

It is unstable because

$$f'(M) = 2AM + \frac{2B}{M^3}, \qquad f'(M_{\rm crit}) = 4AM_{\rm crit} > 0. \tag{5}$$

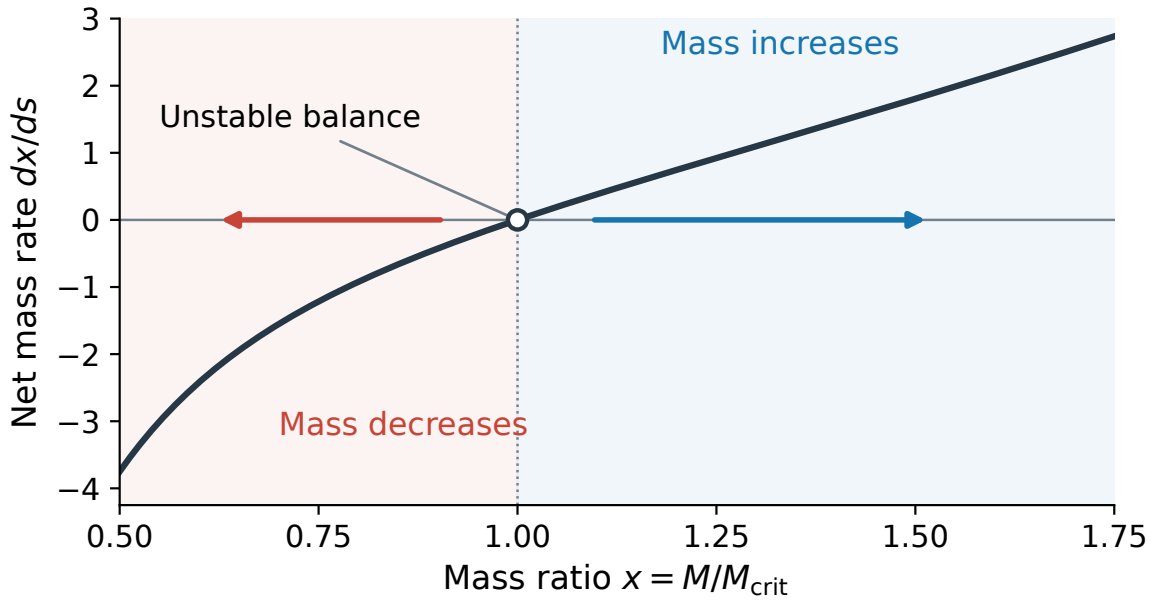


Figure 1: Equal rates do not restore the mass. The net dimensionless mass rate crosses zero at $x = 1$ with positive slope. Arrows show that masses on either side evolve away from the balance. The curve describes every $A, B > 0$ in the fixed-coefficient model.

A small increase in mass strengthens accretion and weakens evaporation, producing further growth; a decrease has the reverse effect. For $M = M_{\rm crit} + \delta M$ with $|\delta M| \ll M_{\rm crit}$, linearization gives

$$\delta M(t) \simeq \delta M(0) e^{t/t_e}, \qquad t_e = \frac{1}{4AM_{\rm crit}}.$$

The balance therefore repels small perturbations, although for weak accretion the e-folding time t_e is long.

3.2 One dimensionless equation

We measure mass in units of $M_{\rm crit}$ and time in units of t_*:

$$x = \frac{M}{M_{\rm crit}}, \qquad s = \frac{t}{t_*}, \qquad t_* = \frac{1}{AM_{\rm crit}}.$$

The equality $B = AM_{\rm crit}^4$ then reduces every positive constant-coefficient case to

$$\boxed{\frac{dx}{ds} = x^2 - x^{-2}}. \tag{6}$$

The vertical axis in Figure 1 is $dx/ds = \dot{M}/(AM_{\rm crit}^2)$; both axes are dimensionless.

3.3 Exact evolution and terminal times

Let $x_0 = M(0)/M_{\rm crit}$ denote the initial mass ratio. Separating Eq. (6) gives $ds = x^2 dx/(x^4 - 1)$. Partial fractions provide a convenient antiderivative:

$$\frac{x^2}{x^4 - 1} = \frac{1}{2}\left(\frac{1}{x^2 - 1} + \frac{1}{x^2 + 1}\right), \qquad F(x) = \frac{1}{4}\ln\left|\frac{x-1}{x+1}\right| + \frac{1}{2}\arctan x.$$

Integrating from x_0 to the current ratio x gives the exact implicit trajectory

$$\boxed{s = F(x) - F(x_0)}. \tag{7}$$

For $0 < x_0 < 1$, this formula applies on $0 < x < 1$ and the mass decreases. For $x_0 > 1$, it applies on $x > 1$ and the mass increases. The exact equilibrium trajectory is $x(s) = 1$. The logarithm in

F tends to $-\infty$ as $x \to 1$ from either side: a nonconstant trajectory cannot reach or cross $x = 1$ in finite time.

For the subcritical branch, $F(0) = 0$. We define $t_{\rm evap}$ as the model's formal time to $M = 0$. Because $F(x) = [\arctan x - \operatorname{artanh} x]/2$ on $0 < x < 1$,

$$\frac{t_{\rm evap}}{t_*} = -F(x_0) = \frac{1}{2}\left[\operatorname{artanh} x_0 - \arctan x_0\right] = \frac{x_0^3}{3} + \frac{x_0^7}{7} + \frac{x_0^{11}}{11} + \cdots . \tag{8}$$

The series is for $0 < x_0 < 1$. Define the pure-evaporation lifetime at the same starting mass as $\tau_{\rm bb}(M(0)) = M(0)^3/(3B) = t_* x_0^3/3$. Its associated trajectory is

$$M(t) = \left[M(0)^3 - 3Bt\right]^{1/3}, \qquad 0 \le t \le \tau_{\rm bb}(M(0)).$$

Equation (8) shows explicitly how a weak accretion term delays that lifetime:

$$\frac{t_{\rm evap}}{\tau_{\rm bb}(M(0))} = 1 + \frac{3}{7}x_0^4 + O(x_0^8) \quad (x_0 \ll 1). \tag{9}$$

For the supercritical branch, $F(x) \to \pi/4$ as $x \to \infty$. We call t_∞ the formal divergence time and $t_{\rm double}$ the time to go from x_0 to $2x_0$:

$$\frac{t_\infty}{t_*} = \frac{\pi}{4} - F(x_0), \qquad \frac{t_{\rm double}}{t_*} = F(2x_0) - F(x_0), \quad x_0 > 1.$$

Both are endpoints of the equation: a real medium has a finite supply of matter, and the semiclassical description fails before $M = 0$. On either side of equilibrium, terminal times grow as $-t_* \ln|x_0 - 1|/4 + O(t_*)$ when $x_0 \to 1$. Exactly at $x_0 = 1$ the mass stays fixed, so there is no finite doubling time.

4 Numerical verification and reference scales

4.1 Method and consistency checks

We numerically integrate Eq. (6) with Python's `scipy.integrate.solve_ivp` using DOP853, relative tolerance 10^{-10}, absolute tolerance 10^{-12} and a maximum dimensionless step of 0.02. To avoid the singular right-hand side near $x = 0$, we use $u = x^3$ for decreasing trajectories; to avoid rapidly growing x, we use $v = 1/x$ for increasing trajectories. These variables obey

$$\frac{du}{ds} = 3(u^{4/3} - 1), \qquad \frac{dv}{ds} = v^4 - 1,$$

respectively. We stop trajectories at $x = 0.05$ or $x = 2.2$ using event detection.

For the four plotted curves, the maximum absolute residual $|s - F(x) + F(x_0)|$ at 501 samples per trajectory is 1.67×10^{-12}. The largest difference between a numerical cutoff time and the exact time in Eq. (7) is 4.92×10^{-13}. The supplied script regenerates the figures and reports the residuals.

4.2 Three distinct timescales

For the reference coefficients,

$$M_{\rm crit} = 3.13 \times 10^{10}\,{\rm kg}, \qquad t_* = 2.44 \times 10^{8}\,{\rm yr}, \qquad t_e = 6.11 \times 10^{7}\,{\rm yr}.$$

By contrast, $M_0/M_{\rm crit} = 3.20 \times 10^{-4}$ and $\tau_{\rm bb}(M_0) = 23.37\,{\rm h}$. The reference hole lies far below the balance mass, and its short lifetime does not carry over to holes that start near $M_{\rm crit}$.

For example, a starting ratio $x_0 = 0.95$ has a formal evaporation time $0.5360 t_* = 1.31 \times 10^8$ yr. A trajectory starting at $x_0 = 1.05$ takes $0.8277 t_* = 2.02 \times 10^8$ yr to double. Both times depend on x_0; t_* is a time unit, not a doubling time. At M_0, Eq. (9) predicts a fractional lifetime correction of only 4.48×10^{-15}; this value follows from the series and lies far below the tolerance of the numerical integration.

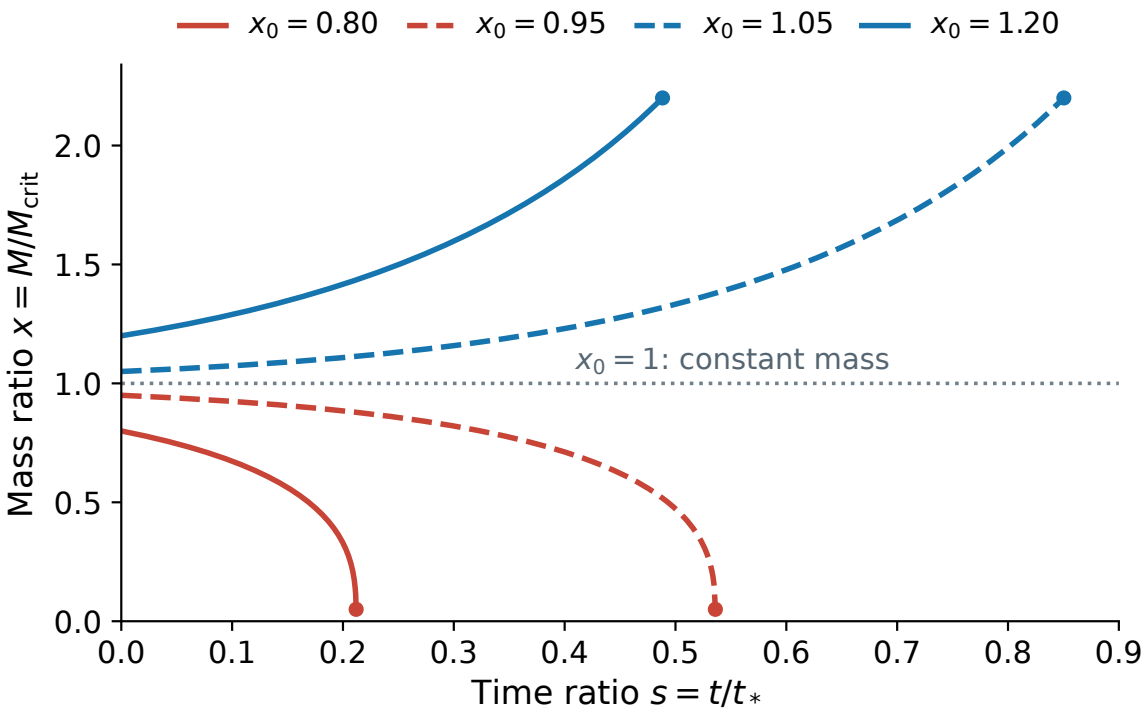


Figure 2: Perturbations move away from the balance on both sides. Four numerical trajectories share a dimensionless time axis. Red curves start below $M_{\rm crit}$; blue curves start above it. Dashed curves begin closer to equilibrium and the dotted line is the exact stationary solution. Filled dots mark the plotting cutoffs. For the reference coefficients, one unit of s is $t_* = 2.44 \times 10^8$ yr, which is distinct from the lifetime at M_0.

5 Physical interpretation and limitations

5.1 Changing the medium shifts an unstable balance

The balance depends on the medium as

$$M_{\rm crit} = \left(\frac{B c_\infty^3}{4\pi \lambda \rho_\infty G^2} \right)^{1/4}.$$

At fixed B and λ, changing ρ_∞ by a factor R changes $M_{\rm crit}$ by $R^{-1/4}$; changing c_∞ by a factor S changes it by $S^{3/4}$. Neither change makes $f'(M_{\rm crit})$ negative.

Let $A_{\rm ref}$ denote the coefficient of our illustrative medium, and let $Q = A/A_{\rm ref}$ while B stays fixed. Equality between the balance mass and M_0 would require

$$Q_{\rm bal} = \frac{B}{A_{\rm ref} M_0^4} = 9.56 \times 10^{13}.$$

If density alone changed while sound speed and λ were artificially held fixed, this would correspond to $\rho_{\rm req} = 9.56 \times 10^{16}\,{\rm kg\,m^{-3}}$. No known material combines this density with that sound speed, and the Bondi model would not apply to one that did. Figure 3 therefore shows the sensitivity of the balance to A, not a range of realizable media.

5.2 The Eddington force balance is a different condition

For an outward luminosity L, suppose the material has an effective flux-mean opacity κ_F. The idealized outward radiative acceleration is $a_{\rm rad} = \kappa_F L/(4\pi r^2 c)$. Equating it to the inward gravitational acceleration GM/r^2 defines

$$L_E = \frac{4\pi G M c}{\kappa_F}.$$

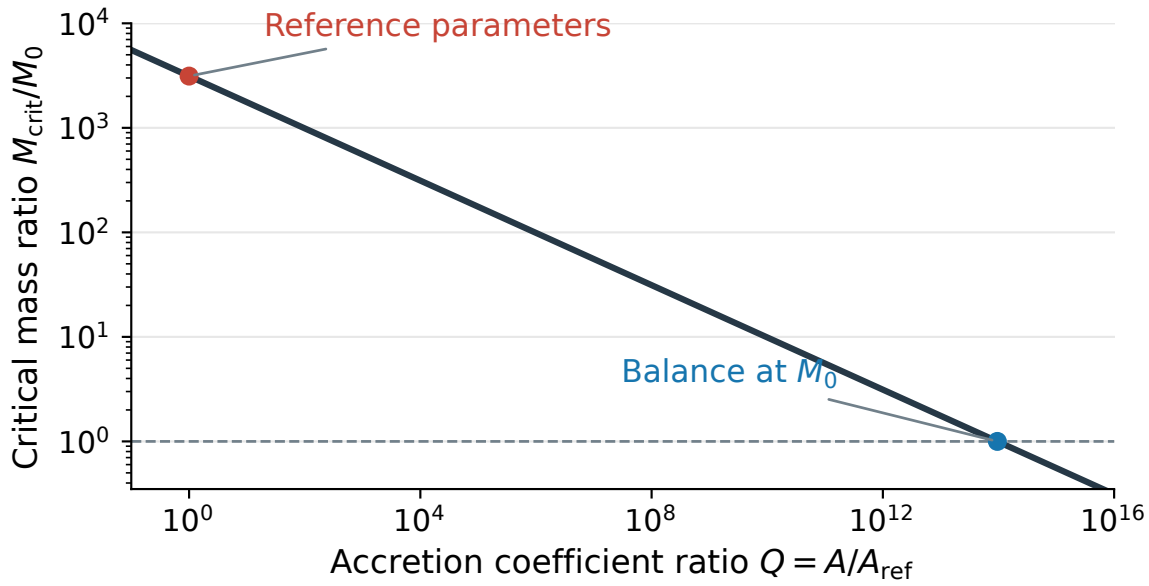


Figure 3: Increasing accretion strength moves, but never stabilizes, the balance. At fixed B, $M_{\rm crit}/M_0 = 3126.65\,Q^{-1/4}$. The reference point is $Q = 1$ and equality $M_{\rm crit} = M_0$ occurs at $Q = 9.56 \times 10^{13}$. Every point on the curve has the positive derivative in Eq. (5).

If we insert $L = K/M^2$ from Eq. (2), the formal force-balance mass M_E obeys

$$M_E = \left(\frac{\kappa_F K}{4\pi G c}\right)^{1/3}, \qquad \frac{a_{\rm rad}}{a_{\rm grav}} = \left(\frac{M_E}{M}\right)^3.$$

This cube-root balance depends on opacity; the fourth-root mass balance in Eq. (4) depends on the accretion coefficient. They are not generally equal. As a formal example, the low-energy Thomson opacity of ionized hydrogen $\kappa_F = \sigma_T/m_p$ yields $M_E = 3.83 \times 10^{10}\,\mathrm{kg}$ and a force ratio of 5.63×10^{10} at M_0. Using this opacity unchanged for a TeV-scale, multispecies Hawking spectrum is unjustified. The coupling fraction and the medium's frequency-dependent absorption and scattering require a separate calculation [8].

5.3 What carries over to other models?

The unstable equilibrium survives any *constant* positive rescaling of A or B. For example, replacing B by ηB with fixed $\eta > 0$ changes $M_{\rm crit}$ by $\eta^{1/4}$ and t_e by $\eta^{-1/4}$ at fixed A; at a fixed initial mass the pure-evaporation lifetime scales as η^{-1}.

A general local test shows where this conclusion stops. Write $\dot{M} = C(M) - E(M)$ with positive gain and loss rates. Suppose they balance at M_* with $C(M_*) = E(M_*) = R_* > 0$. Define logarithmic slopes $p = d\ln C/d\ln M$ and $q = d\ln E/d\ln M$ at M_*. Then

$$\left.\frac{d\dot{M}}{dM}\right|_{M_*} = \frac{R_*}{M_*}(p - q).$$

Here $p = 2$ and $q = -2$. Other laws can give a different sign: $p < q$ gives local asymptotic stability, while $p = q$ needs higher-order analysis. If the gas temperature, density or black hole spin also evolve, the stability of the full coupled system has to be tested separately.

The main physical uncertainties are the continuum Bondi approximation at the reference scale, radiation-dependent heating and flow, realistic Hawking spectra, finite reservoirs and additional degrees of freedom such as angular momentum or charge. None is contained in Eq. (3). Finally, M_0c^2 measures the energy radiated by the evaporating reference hole, not the useful energy, since collection efficiency and weakly interacting emission are not modeled.

6 Conclusion

Within the fixed-coefficient law $\dot{M} = AM^2 - B/M^2$ for positive A and B, there is one positive balance and it is unstable. The exact solution and independently integrated trajectories show motion away from it on both sides. The model also distinguishes a short blackbody lifetime far below equilibrium from a slow departure when the initial mass lies very close to equilibrium.

Passive mass balance is therefore unstable in this idealized Hawking–Bondi equation. The natural next step is to let the radiation change the surrounding medium and hence the accretion rate. A companion paper [7] carries out this extension with a one-zone thermal reservoir and finds that heating shifts the balance but leaves it unstable. Including rotation would require evolving the angular momentum together with the mass.

Acknowledgments

I thank Salizhan Kylychbekov (National Physics Olympiad Committee of the Kyrgyz Republic) for inviting me to write problems for the IPhO-2026 team selection; one of them became the starting point of this study. I am also grateful to Hadi Erturk (University of Oxford) for his mentorship, patience and many helpful discussions throughout this project.

A The coefficient in the Hawking temperature

We use regular Euclidean Schwarzschild geometry to recover the coefficient in Eq. (1) [2, 3]. The Lorentzian line element, in a conventional exterior coordinate chart, is

$$d\ell^2 = -f(r)c^2dt^2 + \frac{dr^2}{f(r)} + r^2d\Omega^2, \quad f(r) = 1 - \frac{r_s}{r}, \quad r_s = \frac{2GM}{c^2}.$$

After continuing $t = -it_E$, set $r = r_s + \epsilon$ near the horizon and define $R = 2\sqrt{r_s\epsilon}$. To leading order in ϵ/r_s,

$$f \simeq \frac{\epsilon}{r_s} = \frac{R^2}{4r_s^2}, \qquad \frac{dr^2}{f} \simeq dR^2, \qquad d\ell_E^2 \simeq dR^2 + R^2d\theta^2, \quad \theta = \frac{ct_E}{2r_s}.$$

For this plane to be smooth at $R = 0$, θ must have period 2π. Hence t_E has period $\Delta t_E = 4\pi r_s/c = 8\pi GM/c^3$. Imaginary-time evolution $e^{-H\Delta t_E/\hbar}$ has the thermal period $\Delta t_E = \hbar/(k_BT_H)$, yielding

$$T_H = \frac{\hbar}{k_B\Delta t_E} = \frac{\hbar c^3}{8\pi GMk_B}.$$

This argument fixes the temperature of the regular thermal Schwarzschild state. On its own it does not calculate outgoing flux after collapse or transmission through the exterior potential; those need a quantum-field and greybody calculation [1, 4].

B Reference values and reproducibility

We use SI units with $G = 6.67430 \times 10^{-11}\,\mathrm{m^3kg^{-1}s^{-2}}$, $c = 299792458\,\mathrm{m\,s^{-1}}$, $h = 6.62607015 \times 10^{-34}\,\mathrm{J\,s}$, $k_B = 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$, $\hbar = h/(2\pi)$, and one year of 365.25 days. The medium parameters are defined in Sec. 2.2. Displayed quantities are rounded; calculations use full floating-point precision.

The accompanying repository contains `main.tex`, `reproduce.py`, three vector figures in `figures/` and `results.json`. Running `python3 reproduce.py` regenerates the figures, constants and integration checks. Running `pdflatex main.tex` twice rebuilds this paper.

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Table 1: Reference values in the stated blackbody and fixed-medium model. The evaporation rate is a positive magnitude.

Quantity	Value	Unit
Initial mass M_0	1.00×10^{7}	kg
Horizon radius $r_s(M_0)$	1.49×10^{-20}	m
Temperature $T_H(M_0)$	1.23×10^{16}	K
Blackbody luminosity $P_{\mathrm{bb}}(M_0)$	3.56×10^{18}	W
Pure-evaporation lifetime $\tau_{\mathrm{bb}}(M_0)$	23.37	h
Bondi scale $r_B(M_0)$	2.97×10^{-10}	m
Accretion coefficient A	4.15×10^{-27}	$\mathrm{kg^{-1}s^{-1}}$
Evaporation coefficient B	3.96×10^{15}	$\mathrm{kg^{3}s^{-1}}$
Accretion rate AM_0^2	4.15×10^{-13}	$\mathrm{kg\,s^{-1}}$
Evaporation rate B/M_0^2	39.6	$\mathrm{kg\,s^{-1}}$
Critical mass M_{crit}	3.13×10^{10}	kg
Time scale t_*	2.44×10^{8}	yr
Perturbation e-folding time t_e	6.11×10^{7}	yr

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